1 The Cluster Decomposition Principle

Problems

1. Define generating functionals for the S-matrix and its connected part:

$$F[v] \equiv 1 + \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int v^*(q_1') \cdots v^*(q_N') v(q_1) \cdots v(q_M)$$
$$\times S_{q_1' \cdots q_N',\, q_1 \cdots q_M}\, dq_1' \cdots dq_N'\, dq_1 \cdots dq_M ,$$
$$F^C[v] \equiv \sum_{N=1}^{\infty} \sum_{M=1}^{\infty} \frac{1}{N!M!} \int v^*(q_1') \cdots v^*(q_N') v(q_1) \cdots v(q_M)$$
$$\times S^C_{q_1' \cdots q_N',\, q_1 \cdots q_M}\, dq_1' \cdots dq_N'\, dq_1 \cdots dq_M .$$

   Derive a formula relating $F[v]$ and $F^C[v]$. (You may consider the purely bosonic case.)

2. Consider an interaction

$$H_{\text{int}} = g \int d^3\mathbf{p}_1\, d^3\mathbf{p}_2\, d^3\mathbf{p}_3\, d^3\mathbf{p}_4\, \delta^3(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}_3 - \mathbf{p}_4)$$
$$\times a^\dagger(\mathbf{p}_1) a^\dagger(\mathbf{p}_2) a(\mathbf{p}_3) a(\mathbf{p}_4),$$

   where g is a real constant and $a(\mathbf{p})$ is the annihilation operator of a spinless boson of mass $M > 0$. Use perturbation theory to calculate the S-matrix element for scattering of these particles in the center-of-mass frame to order g^2. What is the corresponding differential cross-section?

3. A coherent state $|\lambda\rangle$ is defined to be an eigenstate of the annihilation operators $a(q)$ with eigenvalues $\lambda(q)$. Construct such a state as a superposition of the multi-particle states $|q_1, q_2, \ldots, q_N\rangle$.

References

1. The cluster decomposition principle seems to have been first stated explicitly in quantum field theory by E. H. Wichmann and J. H. Crichton, *Phys. Rev.* **132**, 2788 (1963).

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5. N. M. Hugenholtz, *Physica* **23**, 481 (1957).

6. See, e.g., R. Kubo, *J. Math. Phys.* **4**, 174 (1963).

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